TCV-ϕ V: cohesive quantum microstructure from CC-Φ_1 and a controlled bridge to CC-Φ_2 flavour templates

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(Dated: March 12, 2026)

This fifth paper develops the quantum-microstructure layer of the canonical TCV-ϕ programme. Our scope is to formulate a reproducible CC-Φ_1 cluster framework, quantify its curvature response, and connect it to coarse-grained effective quantities used in previous papers. For the flavour bridge, we now adopt a twisted-torus cluster toy as the main exploratory benchmark and compare it against other candidate geometries under a common scoring and naturalness protocol. No complete theory-level fit is claimed. All claims are restricted to script-backed diagnostics archived in the Paper V workspace.

I. INTRODUCTION

Paper V follows Papers I–IV and keeps the same canonical EFT field content and notation. The focus is the microscopic cohesive layer (CC-Φ_1) and its controlled link to exploratory CC-Φ_2 flavour templates. In this revision, the flavour subsection is organized around geometry selection: we compare several cluster topologies with the same PMNS-oriented criteria and then use the best-performing candidate as the primary exploratory benchmark.

II. CANONICAL EFT BASELINE

We keep exactly the same action and parameter hierarchy as in Papers I–IV. No new fundamental fields or couplings are introduced in this paper.

III. LOCAL CC-Φ_1 CLUSTERS: SETUP

We adopt a minimal cluster quantization toy consistent with the CC-Φ_1 preprint construction of Ref. [5]:

$$L_c(R) \sim \frac{1}{m_{\text{eff}}(R)}, \qquad m_{\text{eff}}^2(R) = m_1^2 + \xi_0 R, \tag{1}$$

and

$$\Xi_n(R) = m_{\text{eff}}^2(R) + \left(\frac{n\pi}{L_c(R)}\right)^2. \tag{2}$$

Here Ξ_n has mass-squared units, while the corresponding mode energy scale is

$$\omega_n(R) = \sqrt{\Xi_n(R)}. \tag{3}$$

The baseline benchmark used by `compute_ccphi1_spectrum.py` is $m_1 = 10^{-8}$, $\xi_0 = 0.1$, and $n_{\text{modes}} = 3$.

IV. CC-Φ_1 SPECTRUM AND CURVATURE RESPONSE

From `ccphi1_spectrum.json`, the baseline values at $R = 0$ are

$$\Xi_1 \simeq 1.09 \times 10^{-15}, \qquad \Xi_2 - \Xi_1 \simeq 2.96 \times 10^{-15}, \tag{4}$$

which corresponds to $\omega_1(0) \simeq 3.30 \times 10^{-8}$ and coherence length $L_c(0) \simeq 10^8$ in Planck units. The curvature scan (`compute_ccphi1_curvature_response.py`) uses $0 \leq R \leq 2 \times 10^{-15}$ and yields

$$\frac{\Xi_1(R)}{\Xi_1(0)} \in [1, 3], \qquad \frac{L_c(R)}{L_c(0)} \in [0.577, 1]. \tag{5}$$

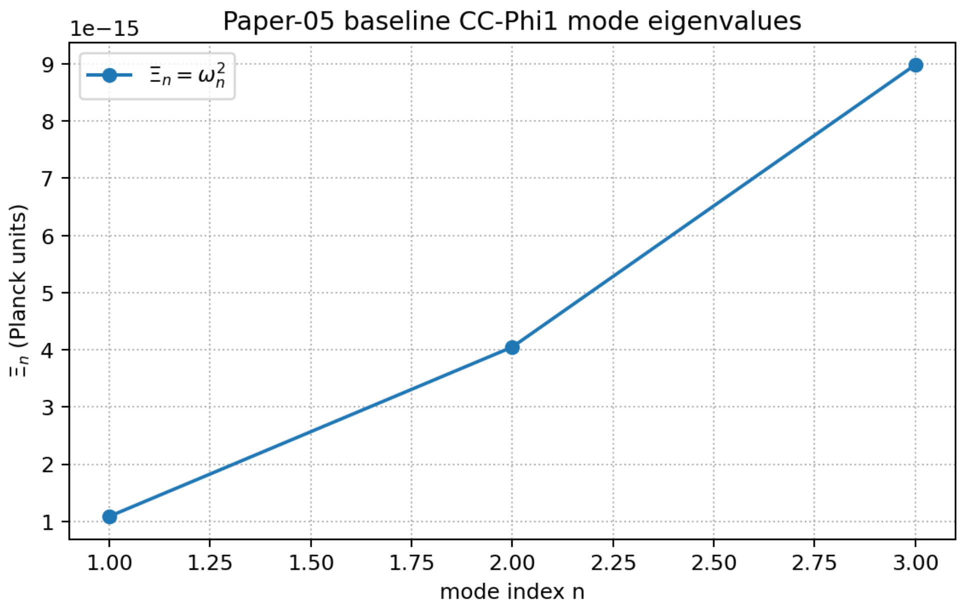

FIG. 1. Baseline CC-Φ_1 mode spectrum from `compute_ccphi1_spectrum.py`.

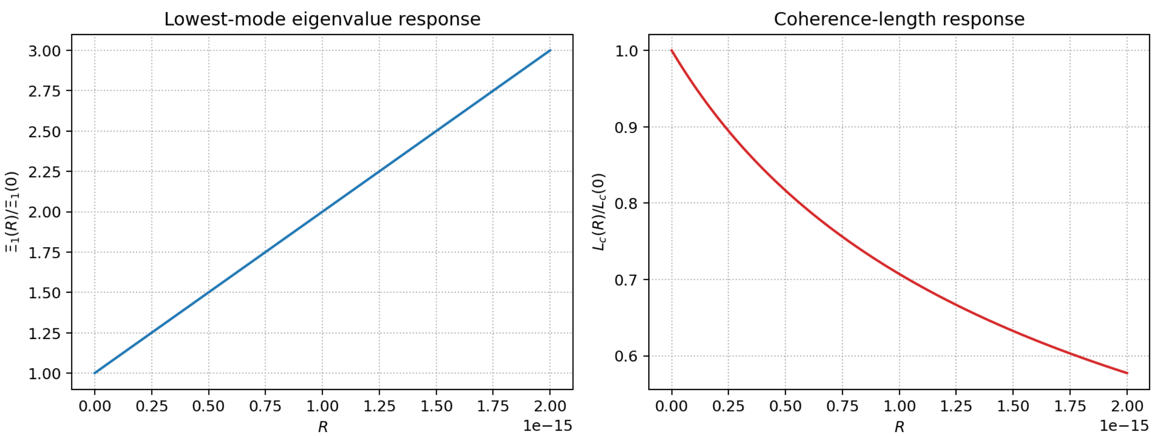

FIG. 2. Curvature response of the lowest mode and coherence length from `compute_ccphi1_curvature_response.py`.

V. COARSE-GRAINING AND EFFECTIVE MACROSCOPIC CONSISTENCY

At the current toy level we use a proxy density $n_{\rm CC} \sim m_1^3$ and define

$$\rho_{\rm cc,eff} \sim n_{\rm CC}\,\omega_1, \qquad p_{\rm proxy} = w_{\rm proxy}\rho_{\rm cc,eff}, \tag{6}$$

with the illustrative closure choice $w_{\rm proxy} = -1/2$ in the current script benchmark. This $w_{\rm proxy}$ is a toy coarse-graining parameter and should not be interpreted as the final EFT vacuum equation of state. The script `compute_ccphi1_coarse_grain.py` gives

$$\rho_{\rm cc,eff} \in [3.30, 5.71] \times 10^{-32}, \tag{7}$$

over the same curvature window. These values are used as consistency diagnostics, not as a final microscopic QFT derivation.

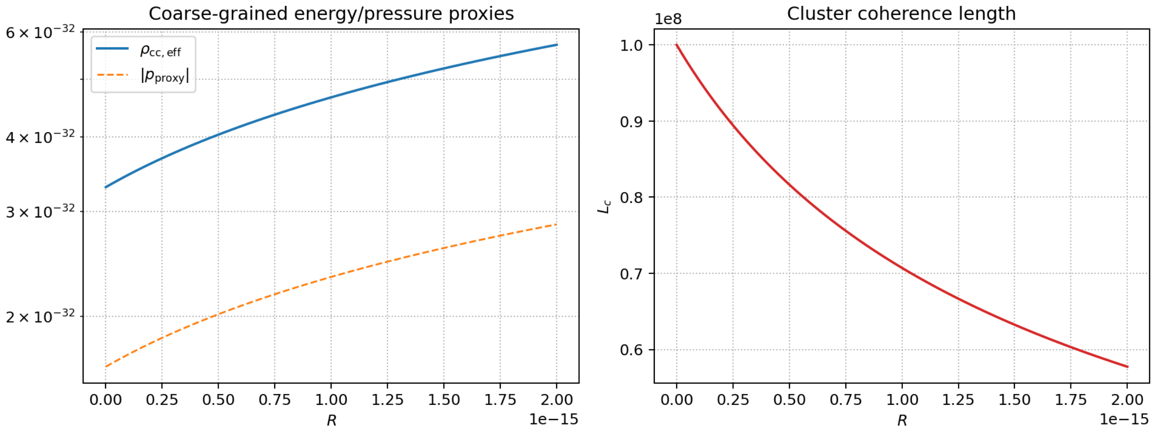

FIG. 3. Toy coarse-graining diagnostics from `compute_ccphi1_coarse_grain.py`.

VI. EXPLORATORY BRIDGE TO CC-Φ_2 FLAVOUR SECTOR: TWISTED-TORUS BENCHMARK

We keep the same conservative status as in previous drafts: this is an exploratory toy layer, not a precision global fit. The central update is geometric. Instead of using a single fixed cluster ansatz, we adopt a twisted-torus (Möbius-ladder) cluster as the primary flavour benchmark, while retaining alternative geometries as comparison points.

The effective flavour extraction remains in the same form,

$$M_\ell = \mathrm{diag}(m_e, m_\mu, m_\tau) + \delta M_\ell, \qquad M_\nu = \mathrm{diag}(m_{\nu_1}, m_{\nu_2}, m_{\nu_3}) + \delta M_\nu, \tag{8}$$

with

$$U_{\mathrm{PMNS}} = U_\ell^\dagger U_\nu. \tag{9}$$

The key difference is that δM_ℓ and δM_ν are now generated from twisted-torus mode structure in `compute_twisted_torus_flavor_` rather than from a single pre-selected pyramidal template.

For the best script-backed twisted-torus benchmark we obtain

$$\theta_{12} \simeq 34.6^\circ, \quad \theta_{13} \simeq 8.29^\circ, \quad \theta_{23} \simeq 41.6^\circ, \tag{10}$$

with a common PMNS proximity score $S_{\mathrm{PMNS}} \simeq 0.52$ (lower is better in our normalization). The PMNS reference point used in this toy score is $\theta_{12}^{\mathrm{ref}} = 33.0^\circ$, $\theta_{13}^{\mathrm{ref}} = 8.6^\circ$, $\theta_{23}^{\mathrm{ref}} = 45.0^\circ$, consistent with current global-fit central values ([7]). In the same scan,

$$n_{\mathrm{strict}} = 82, \qquad n_{\mathrm{relaxed}} = 183, \tag{11}$$

for $n_{\mathrm{natural}} = 4223$ accepted points. The best twisted-torus point lies inside the strict θ_{23} window but not exactly at maximal mixing; this is kept as an exploratory intermediate result. This remains an exploratory indication of geometric viability, not a final flavour solution.

VII. GEOMETRIC EXPLORATION AND MODEL SELECTION DIAGNOSTICS

To avoid hidden fine tuning, we compare candidate geometries under a common protocol: same angle windows, same naturalness bounds, and the same normalized PMNS score used across all toys. The aggregate diagnostics are produced by `compute_flavour_geometry_comparison.py`. The score is defined as

$$S_{\mathrm{PMNS}} = \sqrt{\left(\frac{\theta_{12} - 33^\circ}{10^\circ}\right)^2 + \left(\frac{\theta_{13} - 8.6^\circ}{4^\circ}\right)^2 + \left(\frac{\theta_{23} - 45^\circ}{7^\circ}\right)^2}, \tag{12}$$

and is combined with strict-yield fractions into a PMNS viability index in the summary JSON.

The main conclusion of this section is limited and explicit: within the current exploratory setup, topologically non-trivial geometries (torus ring and twisted torus) produce non-zero strict-window populations in large scans, while the tested rigid geometries (pyramidal, tetrahedral, spherical) remain at zero strict hits. The twisted-torus candidate is the closest tested structure to the PMNS target window without aggressive parameter forcing.

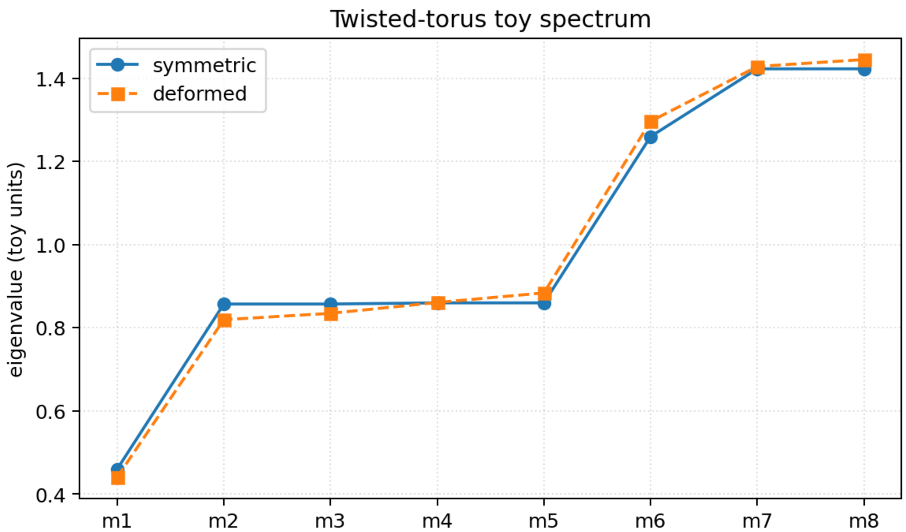

FIG. 4. Twisted-torus (Möbius-ladder) toy spectrum from `compute_twisted_torus_flavor_toy.py`.

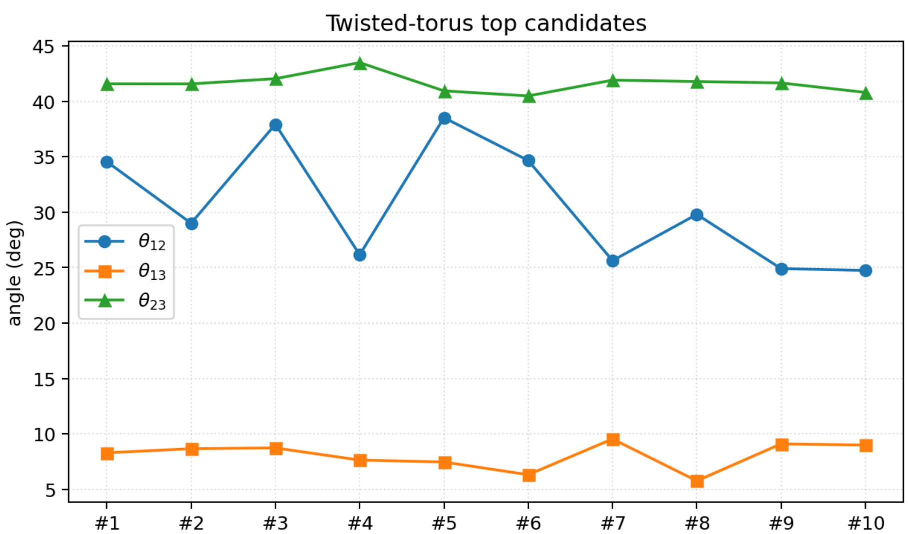

FIG. 5. Top twisted-torus flavour candidates (script-backed).

VIII. SECTOR-COMPATIBILITY STRESS TESTS FOR TWISTED TORUS

We then test whether the same twisted-torus geometry remains compatible with a hierarchical charged-lepton sector and with a weak-mixing quark-like toy sector. The diagnostics are generated by `compute_twisted_torus_sector_compatibility`

For charged leptons, perturbing the charged-sector controls around the best twisted-torus point yields

$$f^{(\ell)}_{\text{strict}} \simeq 0.70, \tag{13}$$

i.e. approximately 70% of tested points remain in the strict PMNS window. For the quark-like toy layer, the mean mixing angles stay small,

$$\langle\theta^q_{12}\rangle \simeq 0.31^\circ, \quad \langle\theta^q_{13}\rangle \simeq 1.7\times10^{-3\,\circ}, \quad \langle\theta^q_{23}\rangle \simeq 2.3\times10^{-3\,\circ}. \tag{14}$$

These tests support compatibility at the toy level, but do not constitute a complete SM flavour derivation.

Geometry	Representative score S_{PMNS}	Strict yield within natural scan	PMNS viability index
Pyramidal	2.40	— (single benchmark)	0.067
Tetrahedral apex/base	2.23	0/1500	0.107
Tetrahedral star	2.64	0/2701	0.013
Torus ring	0.78	59/4358	0.448
Spherical shell	2.69	0/4348	0.000
Twisted torus	0.52	82/4223	0.510

TABLE I. Cross-geometry exploratory comparison from `flavour_geometry_comparison_summary.json`.

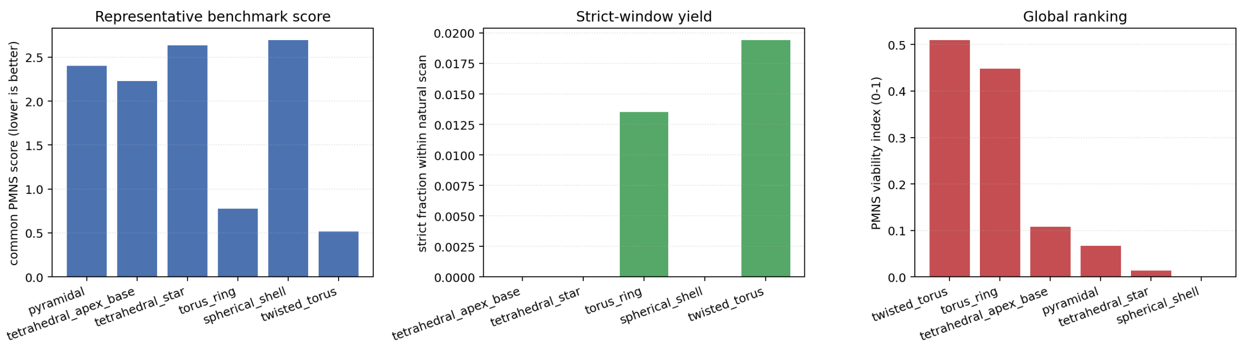

FIG. 6. Common-score and strict-yield comparison across explored cluster geometries.

IX. STATUS OF ASSUMPTIONS

- EFT-derived baseline definitions inherited from Papers I–IV.
- Phenomenological closures used in the CC-Φ_1 cluster pipeline.
- Exploratory geometry-to-flavour mapping for CC-Φ_2.
- Twisted-torus selected as the best current exploratory candidate, without changing the canonical EFT.

X. CONCLUSION

Paper V is designed as a reproducible and conservative microstructure step in the TCV-ϕ series. Relative to earlier drafts, the flavour bridge is now organized as a geometry-selection problem: we test multiple cluster topologies under common criteria and identify twisted torus as the strongest current exploratory candidate. This improves the naturalness narrative at the toy level while keeping all macro-level sectors (bounce, black holes, and cosmological fits) unchanged in this paper. A full micro-to-macro closure from twisted-torus structure is left for future work.

[1] C. Lecroq, "TCV-ϕ I," preprint (2026).
[2] C. Lecroq, "TCV-ϕ II," preprint (2026).
[3] C. Lecroq, "TCV-ϕ III," preprint (2026).
[4] C. Lecroq, "TCV-ϕ IV," preprint (2026).
[5] C. Lecroq, "Cohesive Quantum Gravity and CC-Φ_1 microstructure," preprint (2026).
[6] C. Lecroq, "From Quantum CC-Φ_2 to flavour structures," preprint (2026).
[7] I. Esteban *et al.*, "NuFIT 5.x global fit of neutrino oscillation parameters," online resource (accessed 2026), https://www.nu-fit.org/.

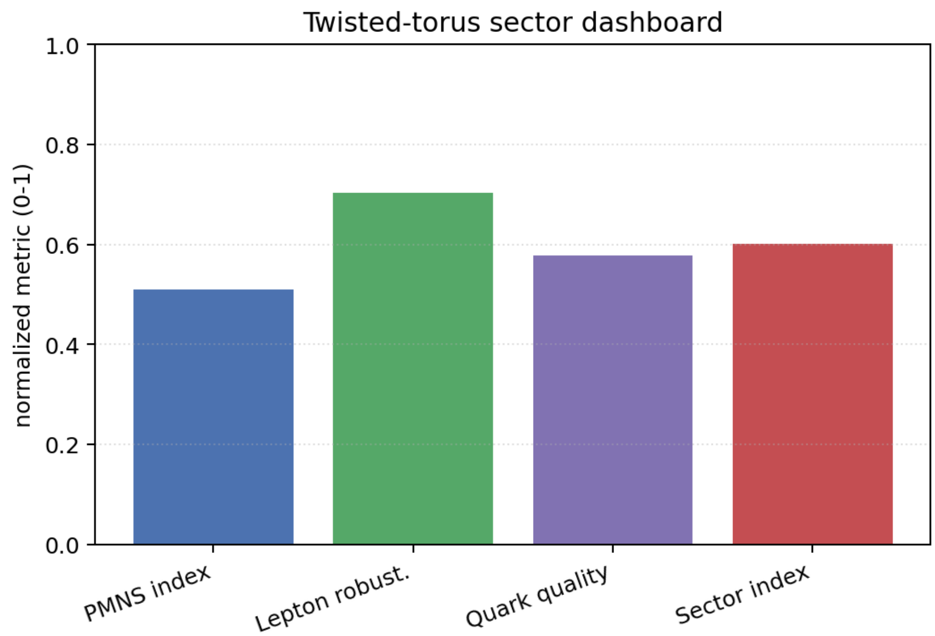

FIG. 7. Twisted-torus sector dashboard: PMNS viability, lepton robustness and quark-like compatibility proxies.

Block	Output artifact	Status label
CC-Φ_1 spectrum	`ccphi1_spectrum.json`	phenomenological toy
Curvature response	`ccphi1_curvature_scan.npz`	phenomenological scan
Coarse-graining	`ccphi1_coarse_grain.json`	toy proxy
Legacy pyramidal flavour toy	`ccphi2_flavour_toy.json`	exploratory legacy benchmark
Twisted-torus flavour toy	`twisted_torus_flavour_toy_summary.json`	exploratory toy
Geometry comparison	`flavour_geometry_comparison_summary.json`	exploratory selection diagnostic
Sector compatibility	`twisted_torus_sector_compatibility_summary.json`	exploratory stress test

TABLE II. Paper-V script-backed outputs and status labels.